

CLOUD COMPUTING

1. What is Cloud Computing?

Cloud computing is the delivery of computing services — including servers, storage, databases, networking, software, analytics, and intelligence — over the internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale. Instead of owning and maintaining physical data centers and servers, organizations can access technology services on an as-needed basis from a cloud provider and pay only for what they use.

2. What is Microsoft Azure Cloud?

Microsoft Azure is Microsoft's public cloud computing platform launched in 2010. It provides a broad and ever-expanding set of cloud services across computing, analytics, storage, and networking.

Businesses can use Azure to build, test, deploy, and manage applications and services through Microsoft-managed data centers spread across the globe.

- 200+ services — from virtual machines to AI and IoT
- Global reach — 60+ regions worldwide
- Hybrid support — seamlessly connects on-premises and cloud
- Enterprise-grade — deep integration with Microsoft 365, Teams, Active Directory
- Security & Compliance — strong data protection and global standards
- Pay-as-you-go Pricing — cost-effective and flexible billing

3. What are the Advantages of Cloud Computing?

- Cost savings — eliminates the expense of buying hardware and software and running on-site data centers
- Speed & agility — most services are provisioned in minutes, enabling rapid prototyping and deployment
- Global scale — deliver the right amount of IT resources from the right geographic location instantly
- Productivity — removes time-consuming IT management tasks like hardware setup and software patching
- Performance — major providers run on worldwide networks of secure data centers with regular upgrades
- Security — cloud providers offer a broad set of policies, technologies, and controls that strengthen overall security
- Reliability — data backup, disaster recovery, and business continuity are easier and less expensive

4. Give 5 Use Cases of Cloud Computing

Use Cases of Cloud Computing

1. Web & Mobile Application Hosting

Cloud platforms allow developers to deploy and manage applications without maintaining physical servers. Features like auto-scaling handle traffic spikes efficiently.

Example:

Netflix runs entirely on cloud infrastructure, scaling automatically to serve millions of users. Airbnb also uses cloud services to handle peak booking traffic without downtime.

2. Big Data Analytics

Cloud computing enables organizations to process and analyze large volumes of data using powerful tools without investing in expensive hardware.

Example:

Spotify uses cloud analytics to process billions of streaming events and provide personalized recommendations. Retail companies analyze large datasets to improve inventory and pricing decisions.

3. Disaster Recovery & Backup

Cloud services provide data backup and recovery solutions across multiple geographic locations, ensuring business continuity during failures.

Example:

Using cloud recovery solutions, organizations can restore systems within minutes. Governments and enterprises use cloud backup to protect critical data from disasters.

4. Machine Learning & Artificial Intelligence

Cloud platforms offer high-performance computing resources (like GPUs) and pre-built AI tools for developing intelligent applications.

Example:

Large-scale AI models are trained using cloud infrastructure. Healthcare companies use cloud-based AI to analyze medical data and improve diagnosis accuracy.

5. Collaboration & Remote Work

Cloud-based tools enable real-time collaboration, communication, and file sharing from anywhere.

Example:

Platforms like Microsoft Teams and Google Workspace allow global teams to work together efficiently. Many organizations rely on these tools for remote operations.